

Calibrated Risk Envelopes for LLM Quantization, and the Limits of Predicting Compression Damage

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Abstract

Quantizing a large language model reduces serving cost and costs some accuracy. Determining how much normally requires re-running a full benchmark suite, which is the expense quantization was meant to avoid. We ask whether the accuracy delta can be predicted instead. Using 817 published evaluations spanning 38 base checkpoints, 8 model families and 6 quantization schemes, we report three results. First, a *negative* one: per-model prediction carries almost no signal. Under leave-one-family-out validation the best predictor, a per-scheme mean, beats a global mean by 0.03pp of MAE (95% checkpoint-bootstrap CI [0.01, 0.05]), and both ridge regression and gradient boosting are *worse* than guessing the average, because evaluation noise accounts for at least a third of the variance on 8 of 11 benchmarks. Second, a positive one: conditioning only on the quantization scheme yields per-scheme conformal-style intervals with 90.1% empirical coverage (118/131, 95% CI [83.6, 94.6]) on checkpoints never used in construction. Those checkpoints cover 4 of the 6 schemes, and W4A16 supplies 80 of the 131 rows. Each (scheme, size-band) cell returns one of three verdicts: trusted; insufficient evidence, where the interval is still shown but the scheme is demoted and flagged; or refused, where the interval is withheld. Currently 9 of 17 cells are insufficient evidence and 0 are refused. Third, and most consequential for practice: the published corpus is outcome-truncated. Every configuration in it is one that someone chose to release. We rented an A100 and measured deliberately-faulted configurations, observing losses to -39.5pp against a published worst case of -8.86pp —measured, we stress, only on models of 1.5B and below, and on configurations engineered to fail rather than typical ones. This is an existence proof that far worse outcomes exist than the corpus contains; it does not estimate how common they are. We argue the resulting artifact should be described as a calibrated historical baseline rather than a predictor, and we state precisely why its coverage guarantee does not transfer to a practitioner’s own untuned recipe. Code, data and the provenance of every reported figure are provided as supplementary material.

1 Introduction

Post-training quantization is now routine. A practitioner choosing between W4A16, W8A8 and FP8 wants one number: how much accuracy will this cost? In practice the only way to know is to run the evaluation.

This paper asks whether that number can be forecast from the public record. Model publishers release quantized checkpoints with benchmark tables comparing the compressed model against its dense parent. Aggregated, these are hundreds of paired measurements of exactly the quantity of interest (850 extracted in our harvest).

Our contributions:

1. **A negative result** (§5). Per-model prediction of the accuracy delta does not work, and we

show why: the target is dominated by evaluation noise. We report this as the primary finding rather than burying it.

2. **A calibrated envelope that does work** (§6), conditioned on scheme alone, with measured coverage on held-out families and a three-state verdict for each (scheme, size-band) cell: *trusted*; *insufficient evidence*, where the interval is still shown but the scheme is demoted and the cell flagged as one the tool cannot judge; or *refused*, where the interval is withheld. Currently 9 of 17 cells are insufficient evidence and 0 are refused.
3. **An existence proof that the corpus omits severe outcomes** (§7), via GPU experiments on deliberately-faulted configurations—data the published record structurally cannot contain. It shows that losses far beyond anything published are reachable on small models. It does not estimate how large the bias is: our attempt at a correction (in the supplement) did not work well enough to ship.
4. **An audit** (§9) recording errors we found in our own work, including ones that changed published numbers and claims.

2 Related Work

Predicting evaluations. BenchPress [13] establishes the mechanism closest to ours: predict an eval result, estimate whether to trust it, wrap it in a conformal interval, and decide whether to skip the real run. Their score matrix over 84 frontier models and 133 benchmarks is approximately rank-2. We do not claim novelty against them on mechanism. Their estimand is a model’s absolute score; ours is the paired delta between a model and its compressed self, and quantized checkpoints do not appear in their paper.

Conformal prediction on compressed models. Tong et al. [9] apply conformal prediction directly to quantized and sparse LLMs, which is a closer domain than BenchPress. The estimand differs: they evaluate predictive uncertainty (coverage and set size) *after* running the compressed model. Ours is constructed over historical accuracy deltas to give an interval *before* running anything. Their Figure 2 plots the same quantity we predict, $\text{Acc}_{\text{compressed}} - \text{Acc}_{\text{dense}}$, but measures it. Their finding that compression decouples accuracy from uncertainty bounds what an accuracy-only estimate like ours can claim (§8).

Quantization methods. The checkpoints we harvest are produced with GPTQ [3] and SmoothQuant [12] via llm-compressor [10]. Kumar et al. [6] fit a functional form that predicts degradation from training and inference in varied precisions, using pretraining runs of models up to 1.7B parameters. We fit no such law: our envelope is empirical, per scheme, over published post-training-quantized checkpoints.

The source of our data. Kurtić et al. [7] report over 500,000 evaluations of quantized Llama-3.1 models, finding recovery above 99% of the average OpenLLM v1 score for every scheme tested. As far as we can tell, that work does not address whether the published set of quantized models is a biased sample. Characterising that gap is the contribution of §7.

Selection bias. Survivorship bias has been identified and measured in a widely used retrieval benchmark [4]. Conformal prediction’s guarantee is known to fail under selection: conditional on a selection event, calibration data are no longer exchangeable with a test point [1, 5]. We apply this directly to our own artifact in §8.

3 Data

We harvest published model cards into rows of (model, scheme, benchmark, $\text{acc}_{\text{before}}$, $\text{acc}_{\text{after}}$), yielding 850 rows, of which 817 remain after dropping 33 whose baseline accuracy is below 20% (near chance, where a delta is uninformative), across 38 base checkpoints, 8 families and 6 schemes.

An integrity gate. Cards typically print a *Recovery %* column alongside the before/after pair. This is redundant information, and we exploit it: for every three-column row we check $\text{recovery} \approx 100 \times \text{after}/\text{before}$ to a tolerance derived from the card’s own printed precision, and reject rows that disagree. This caught spurious values that our parser produced under column-order variation between card templates, and internally inconsistent cards. Rejections are recorded rather than silently dropped.

A structural limitation, stated at the outset. All rows come from a single publisher using a single toolchain. Holding out a model family tests transfer across *models*; nothing in this design tests transfer across *quantization practitioners*. §7 gives direct evidence that this distinction matters.

The pathway is only sparsely documented. A live scan on 2026-09-26 across the 439 RedHatAI quantized-model cards currently published on Hugging Face (of which 102 contribute rows to our harvested dataset) shows how sparsely the pathway that produced these outcomes is documented: 12 of 439 cards pin a quantization-library version, 33 mention activation reordering, 17 mention dampening, and 198 of 439 report none of `num_calibration_samples`, `max_sequence_length`, activation reordering or dampening. On the 102-card subset that our dataset draws rows from, 0 pin a library version and 39 of 102 report none of those four parameters. (This RedHatAI-only scan is distinct from the 10-publisher, 260-card publisher census reported in §9, which is an earlier stable snapshot spanning multiple publishers and asks a different question.) The published record we calibrate against is therefore a record of outcomes with the pathway that produced them substantially undocumented on the publisher whose cards contribute the largest single share of our corpus, which is one of the reasons an outcome-truncation argument (§7) cannot be closed from the corpus alone.

4 Method

Conformal-style intervals. We use the residual-quantile construction of split conformal regression [8]: given residuals $\{s_i\}_{i=1}^n$ and level α , the interval half-width is the $\lceil (n+1)(1-\alpha) \rceil$ -th smallest score, which under a proper calibration/test split gives exact finite-sample coverage under exchangeability. Where n is too small for that index to exist, we return ∞ rather than a misleadingly finite band. The shipped per-scheme interval fits its centre and its half-width on the *same* rows (an in-sample residual-quantile band), so the split-conformal guarantee holds only for the explicit calibration-set variant used inside the leave-one-family-out audit; the empirical prospective figure below is what carries the shipped intervals.

Mondrian (group-conditional) calibration. Calibrating separately within categories is the Mondrian construction of Vovk et al. [11], applied to regression by Boström and Johansson [2]. A single marginal interval advertised at 90% covers only 62% of NVFP4 rows under leave-one-family-out (recomputed live)—the users who most need it. We therefore calibrate per scheme.

A three-state verdict. For each (scheme, size-band) cell we measure coverage directly, and the tool returns one of three verdicts. A cell is *trusted* when measured coverage is adequate and adequately supported. It is *refused*—INSUFFICIENT CALIBRATION, no interval—when coverage is measurably poor: the checkpoint-bootstrap interval on its coverage lies wholly below the 85% line, on at least 3 checkpoints. It is *insufficient evidence* when the cell rests on fewer than 3 distinct checkpoints, or when a checkpoint-level cluster bootstrap 90% interval on its coverage straddles the 85% line. In that case the evidence cannot support either of the other two verdicts. The tool still prints the all-sizes interval for such a cell, but demotes the scheme to the lowest tier and states that the cell cannot be judged; only a refused cell has its interval withheld.

Coverage is scored *one-sided*: how often the true delta stayed at or above the interval’s lower bound. The risk this tool bounds is accuracy loss, and a model that beats its envelope has exposed no one to anything, so counting that as a failure would flag cells that are merely outperforming. Comparing a one-sided measurement against the printed 90% is the *lenient* reading, not a flattering one: the interval is symmetric, so the permitted 10% is split across two tails, and pooled one-sided coverage, across all 817 rows with each row scored under 7 different calibration families, is 95.1%.

Currently 0 of 17 cells are refused. Two cells were flagged under the earlier two-state rule, and neither is a confident refusal under the three-state one. `w4a16|<2B` had one-sided coverage of 70.1% over 11 rows (two-sided 68.8%), but those rows come from only 2 checkpoints. `w8a16|>10B` had 81.1% over 31 rows from 5 checkpoints (two-sided 73.7%), and its bootstrap interval, [47, 100]%, straddles the threshold. Both are now *insufficient evidence*. We had scoped the change to those two cells; applying the rule to every cell moved 9 of 17 into that state, including several whose point-estimate coverage looked comfortable. The same measurement and the same rule judge each scheme’s all-sizes coverage: 1 of the 6 schemes is insufficient evidence at that level and 0 is refused. The one-sided scoring and the three-state rule were proposed by an external commenter who identifies as an AI account collaborating with the OpenLanguageModel maintainers; we recomputed each of their figures from the raw rows before adopting either change, and one of their descriptions did not survive that check.

A support floor. Coverage measured on many benchmarks from one checkpoint is not independent evidence; it is one quantization run viewed many ways. Cells resting on fewer than 3 distinct checkpoints cannot reach the top tier regardless of measured coverage. 4 cells are affected.

5 The negative result

Under leave-one-family-out validation, row-weighted MAE:

predictor	MAE (pp)	beats global mean?
per-scheme mean (shipped)	0.7227	yes
global mean (baseline)	0.7545	—
per-(scheme $\times$ benchmark)	0.7607	no
ridge regression	0.7639	no
per-benchmark mean	0.7647	no
gradient boosting	0.7878	no

The per-scheme mean’s advantage is small: 0.03pp, with a checkpoint-bootstrap 95% CI of [0.01, 0.05]pp, and it is positive in 7 of the 8 held-out families. It is a distinguishable but modest gain. Ridge and gradient boosting are both *worse* than predicting the global average. Adding model size, bit-width, method, benchmark identity and headroom each degraded out-of-family performance.

Why. Estimating per-benchmark evaluation noise from near-lossless schemes—whose true delta should be approximately zero—gives a noise floor that accounts for anywhere from 15% of the observed variance to all of it across 11 benchmarks. On GSM8K the deltas of supposedly lossless schemes have a standard deviation of 2.1pp, while the quantity being predicted has mean -0.39 pp. The target is a difference of two noisy quantities that largely cancel: the numerically smallest and hardest-to-predict summary of quantization damage available. Two cautions on the noise figures. They assume near-lossless schemes have a true delta near zero, so they are upper bounds on noise. And 5 of the 11 estimates rest on 10 or fewer rows; one, at 124%, exceeds the total variance, which we attribute to sampling error rather than treat as a finding.

Implication. The entire fitted artifact is 18 numbers. A raw per-scheme historical quantile band, with no model and no conformal machinery, is a serious competitor: it looked better covered in-sample, though under the strict protocol that advantage disappears (§9). What conformal contributes is a finite-sample guarantee and a principled rule for small- n schemes, not accuracy. We therefore describe the result as a *calibrated historical baseline*, not a predictor.

6 Results

What this tool is for. The scenario it answers is: a practitioner is considering a scheme on a checkpoint from a family already represented in the published record, and wants a calibrated range for the accuracy delta with an honest verdict on whether the record supports that range for that cell. Within that scope, each cell returns an interval and one of three tier labels; outside it, the tool declines to speak. It is not a scheme-safety certificate, it does not bound a scheme’s worst case, and it does not extrapolate to recipes or families the corpus has not already seen – §7 and §8 state each of those out-of-scope cases directly.

scheme	90% interval (pp)	worst observed	% losing >3pp	evals
fp8	[-1.11, +0.48]	-1.90pp	0.0%	80
w8a8_int	[-1.71, +1.07]	-3.32pp	1.0%	194
fp8_dynamic	[-1.44, +1.34]	-8.72pp	1.0%	200
w8a16	[-0.98, +0.81]	-3.04pp	1.2%	86
w4a16	[-2.87, +1.40]	-8.86pp	5.7%	193
nvfp4	[-4.10, +1.87]	-8.12pp	15.6%	64

Prospective validation. After freezing the method we harvested checkpoints not used during development. Under a strict protocol that excludes any family present in training and any card whose format informed a parser fix, coverage is $118/131 = 90.1\%$, 95% Clopper–Pearson CI $[83.6, 94.6]$, against a nominal 90%. A from-scratch, standard-library-only reimplementation sharing no code with the pipeline reproduces this figure with zero disagreements. That checks the computation, not the validity of the design.

What the prospective set does and does not cover. The 131 strict rows come from 19 quantized checkpoints, not 131 independent observations: a bootstrap that resamples checkpoints gives a 95% interval of $[83.2, 94.7]\%$, close to the row-level one. The rows cover only 4 of the 6 schemes; FP8 and NVFP4 have no prospective rows, so the figure says nothing about them.

scheme	strict rows	checkpoints	covered (%)
w4a16	80	11	87.5
w8a8_int	36	5	91.7
fp8_dynamic	11	2	100.0
w8a16	4	1	100.0

W4A16, the most common scheme, dominates the pooled figure and sits below the nominal 90%; the other schemes rest on few checkpoints. Over all 186 harvested prospective rows, before the strict filter, coverage is 90.3%, so the filter did not manufacture the headline. That said, the pooled 90.1% is pulled up by 11 FP8-dynamic rows from 2 checkpoints and 4 W8A16 rows from 1 checkpoint, both at 100.0% coverage: W4A16, at 80 of the 131 strict rows (61%), comes in at 87.5%, which is below the nominal 90%. The pooled number should be read as an average across schemes with very different sample sizes, not as a coverage claim uniform across schemes.

What the envelope cannot do. 0 of 6 intervals exclude zero: the tool can never state that a scheme *will* cost accuracy. And 19 of 25 observed losses worse than 3pp fell *below* their interval floor—when real damage occurs, the interval usually misses it. This is the single most important caveat in the paper.

7 The measured left tail

The corpus contains only configurations someone chose to publish. To measure what it structurally cannot show, we rented an A100 and quantized real models under deliberately-faulted recipes.

Result. Worst measured loss -39.5pp , against -8.86pp as the worst loss anywhere in the published corpus. 16 of 36 faulted rows lost 3pp or more; separately, 2 of 9 rows from our correctly-configured control arm did too. The dominant single effect was wrong scale granularity—per-tensor instead of per-group.

Scope, stated plainly. This tail was measured only on models of 1.5B and below. It establishes that catastrophic damage is reachable; it is not a measured bound for larger models.

A confound we found and did not resolve. Re-running MMLU under the published protocol (5-shot, letter-scored) recovers accuracies in the range published on the cards: our Qwen2.5-0.5B-Instruct scores 47.20 against the Instruct card’s 46.83 (the base-model card gives 47.57), and our Qwen2.5-1.5B-Instruct scores 59.57. The harness question closed. Our control run then compared Instruct-model checkpoints against Red Hat’s *base-model* W4A16 publications, because no Instruct W4A16 checkpoints are published: our correctly-configured Instruct control is $2.6\times$ more damaging than the base-model published W4A16 (mean gap -3.78pp, $z = 2.9$ and 5.3). This is not like-for-like—Instruct models could plausibly be more fragile under 4-bit quantization than their base counterparts—so the ratio is indicative rather than a clean estimate. A matched Instruct-vs-Instruct control is deferred future work. The pattern we observed is consistent with approximately $171\times$ less calibration data than production defaults, though the attribution is not independently tested. Consequently, excess-over-control measures *bad recipe versus mediocre recipe*, not versus production. It is an indicative anchor, not a bound, and we decline to publish a corrected point estimate: the correction moves by 5.70pp depending on the anchor set, and we have only 6 rows from two models.

8 Limitations

The guarantee is conditional, and absent where it matters most. Split conformal gives exact coverage under exchangeability between calibration and test. Our calibration set is the set of configurations a publisher chose to release—a selection event. Conditional on selection, calibration points are not exchangeable with an arbitrary test point and the guarantee does not transfer [1, 5]. It holds for a new checkpoint drawn from the same publication process. It carries **no guarantee** for a recipe the reader tuned themselves and that no publisher would have released—which is the case a practitioner most wants covered.

Accuracy is not the whole of deployment risk. Tong et al. [9] show compression frequently decouples accuracy from uncertainty: a model can hold accuracy while its prediction sets inflate. A scheme this work calls low-risk on accuracy may still degrade uncertainty quality. We measure none of that.

One publisher, one toolchain, and why. See §3. Our own control arm (§7) is direct evidence that practitioner and calibration volume can dominate scheme choice. We measured how confining this is: surveying 10 publishers and 260 quantized model cards, 10 of 222 cards from the other 9 publishers carry paired before/after evaluations, 9 of them Intel’s, so others do publish them. What the other publishers almost never provide is a *Recovery* column, which is what lets each row be checked against $100\times$ after/before and rejected on disagreement: of their rows, 6 pass that gate, all from 1 card, against 147 from the control publisher. The corpus is single-publisher not because others publish nothing, but because the control publisher’s cards are, by a wide margin, the ones published in a self-verifying form.

What can and cannot be re-run. The supplement recomputes every tagged figure from the data files it ships. It cannot re-run the faulted-configuration experiments (they need a GPU; the recorded results are included), the harvest from raw model cards (the cards are not redistributed), or the prospective-card checks (those files are not included), and re-running the low-rank comparison mentioned in the supplement needs a checkout of BenchPress’s released code.

An attempt at re-running the calibration-data test. We attempted to reproduce Red Hat’s published W4A16 checkpoints for two of the corpus’s most-measured base models, Qwen2.5-0.5B and Qwen2.5-1.5B, using the `llm-compressor` recipe at its documented defaults, in order to test whether the roughly $171\times$ calibration-data ratio between our own control and production defaults accounts for the observed excess damage. Neither of these two cards pins a library version or names a calibration dataset; both record only the resulting weight representation (symmetric per-group INT4, group size 128 on 1.5B and group size 64 on 0.5B).

Rows are not independent. Up to sixteen benchmark rows come from a single quantization run. We mitigate with group-conditional calibration and the support floor, and note that a bootstrap over the 19 quantized checkpoints gives [83.2, 94.7]%, close to the row-level interval, but the unit of analysis remains a row. The leave-family-out coverage measurement has the same dependence: each row is scored once under each of 7 calibration families, so the 5719 evaluations are not independent, and every coverage judgment in this paper resamples whole checkpoints rather than counting evaluations.

9 Audit

We record errors found in our own work, including those that changed published claims. This section exists because the alternative—reporting only what survived—would misrepresent how the result was obtained.

Errors that changed a published number.

- *A retracted headline.* An early claim that held-out models came from families never used in construction was wrong; they were new checkpoints of families already present. The strict figure is 118/131.
- *Two parser bugs.* A benchmark label typo cost 11 training rows; separately, MATH-500 was matched as `Math-Lvl-5`, contaminating 7 test rows with scores from a different benchmark.
- *A centring mismatch.* Calibration measured residuals about the stratum mean while intervals centred on the scheme mean, mis-sizing every calibrated width. Found only by independent reimplementations.
- *A column-order fault.* One card template orders columns [recovery, base, quant], which our parser turned into a spurious -68pp delta. The integrity gate rejected 10 of 11 rows on that card, passing only the one where base = quant made the check degenerate.
- *A retracted single-publisher claim.* We had asserted that one publisher was close to the only source of paired evaluations. A census of 10 publishers found 10 paired-evaluation cards outside the control publisher, 9 of them Intel’s; what differs is whose rows can be verified (§8).
- *A bootstrap on the wrong statistic.* The checkpoint-level bootstrap in our coverage audit resampled the two-sided indicator while the verdict was judged on the one-sided statistic. An external commenter asked for it to wrap the one-sided figure, which exposed the mismatch (§4).

- *Inflated pair counts.* An earlier version of the tool reported the number of (test row, calibration family) evaluations as if it were the number of rows: with 8 families each row is scored 7 times, so 5719 evaluations came from only 817 rows. Every row-count claim was recomputed on `distinct_rows`.
- *Two levels judged differently.* Cells were judged on the one-sided coverage statistic and refused when the point estimate fell below the threshold, but schemes had been judged on the two-sided statistic and refused only when the entire bootstrap interval fell below it. A single classifier now judges both levels the same way, and the envelope’s refused list is a computed copy of the ranking tool’s, checked by a test.

A regression we nearly shipped. An audit indicated that a raw asymmetric empirical band covered better than the shipped symmetric conformal interval (93.4% versus 89.6%). Since quantization damage is left-skewed, a symmetric band is obviously the wrong shape, and we were about to switch. On measuring first, the empirical band returns an *infinite* interval on 43% of rows, because a two-sided empirical index requires more calibration points than are often available. An infinite interval covers by construction. On rows where both are finite, the original wins on coverage *and* width. The change was reversed.

An overclaimed guarantee. The artifact was described as carrying a coverage guarantee without qualification. Checking the method against the conformal literature rather than against our own earlier work showed the guarantee is conditional on exchangeability that selection breaks. Corrected in §8.

Provenance discipline. Two parser fixes were informed by inspecting rows that were already in the test set. Rather than silently retain them, we record which fixes were test-informed and exclude the affected family from the strict headline.

An attempted extension of the adversarial arm to 3B. On a fresh 40 GB A100 with a fresh dependency install, we re-ran the adversarial sweep to (i) verify that yesterday’s numbers reproduce and (ii) add a Qwen2.5-3B-Instruct row. The regression check passed: base evaluations on Qwen2.5-0.5B-Instruct and SmolLM-135M-Instruct reproduced byte-for-byte, deterministic-quantization recipes (per-tensor RTN, w8a8 without smoothing) reproduced byte-for-byte, and stochastic GPTQ recipes stayed within $\leq 0.85\text{pp}$ of yesterday, consistent with fp16 Hessian-accumulation noise. The 3B extension did not land: base evaluations on Qwen2.5-3B-Instruct completed, but all five quantization recipes hit `torch.cuda.OutOfMemoryError` during the paired-evaluation forward pass at batch size 2, not during quantization itself; the fp16 model weights (≈ 6 GB), PyTorch’s caching allocator fragmentation from unloaded prior states, and activation memory for a batch of 2 long sequences together exceeded the 40 GB budget with roughly 10 GB of free memory needed for the next allocation. A single 80 GB card would be expected to accommodate this, so the constraint is hardware, not methodology. The adversarial arm therefore remains bounded at $\leq 1.5\text{B}$ as before; §8 states this scope directly.

Mechanical verification. The figures this paper reports as results are macros generated from a claims registry that recomputes each value from source artifacts, and the accompanying code fails a test if an outward-facing document contains an untraced numeric literal. The exceptions are historical figures in the error list above (row counts and the -68pp delta), which are quoted from the provenance record rather than recomputed.

10 Conclusion

Per-model prediction of quantization accuracy damage does not work on published data, and the reason is largely measurement noise rather than model capacity. What does work is narrower: a per-scheme calibrated envelope, validated at 90.1% coverage on unseen checkpoints for 4 of the 6 schemes, and that says which of its own cells it cannot judge. Each cell is trusted; or insufficient evidence, in which case the interval is still shown but flagged and the scheme demoted; or refused, in which case the interval is withheld. Today 9 of 17 cells are insufficient evidence and 0 are refused. The more useful contribution may be the limitation itself—the published record of quantized models is outcome-truncated, and we measured, on deliberately faulted recipes for models up to 1.5B, losses more than four times worse than anything it contains. A risk estimate built on that record is a floor on risk, not a ceiling, and should say so.

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